PRANAV RAJESH CHARAKONDALA

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PROFESSIONAL SUMMARY

Software engineer graduating May 2026 with 2+ years of industry experience, now focused on building production LLM platforms, RAG systems, and agentic AI pipelines across AI engineering, automation, and application development, with a strong foundation in Python, Java, distributed systems, and generative AI across enterprise fintech and research environments.

PROFESSIONAL EXPERIENCE

CENTER FOR HEALTH INFORMATICS, UIUC | Champaign, IL

Lead Software Engineer – AI | *Full-Stack AI Platform Engineering* **May 2025 – Present**

- Designed a **provider-agnostic LLM platform** for real-time health misinformation detection across short-form videos, supporting OpenAI, Anthropic, and Ollama through a runtime-checkable Protocol abstraction, enabling **model switching** and extensible integration of AI APIs
- Engineered a **distributed task execution system** using Redis-backed Celery workers with SHA-256 idempotency, tenacity-based exponential backoff, and structured per-job logging, ensuring **reliability and scalability** of AI inference across containerized microservices
- Built an **automated model evaluation framework** benchmarking multiple LLM providers across precision, recall, and F1 per label with YAML configs and confidence histograms across 10+ runs, enabling data-driven provider selection and **prompt version management**
- Integrated **evidence-based PubMed grounding** via NCBI E-utilities API, extracting specific health claims and mapping them to citations, implementing **data quality checks, scalable ingestion frameworks, and observability** for production LLM deployments

DELOITTE USI | Bangalore, India

Analyst – Software Engineer | *Financial Services Platform* **Jun 2022 – Jul 2024**

- Delivered 50+ fullstack features in Java and Node.js for a U.S. wealth management platform supporting 12,000+ financial advisors, optimizing **SQL query performance** by 20% and reducing page load latency to ~2s
- Designed **asynchronous, event-driven workflows** using Platform Events and async callouts across distributed services, reducing cross-system data latency by 40% and improving system responsiveness across enterprise financial workflows
- Led **code reviews** and design sessions across cross-functional UAT and deployment cycles, resolving 60+ integration defects, reducing critical bugs by 30%, mentoring 2 junior analysts, and producing technical documentation to standardize deployment workflows
- Automated resolution of 200+ Salesforce–DB2 data discrepancies using **SQL and Snowflake-based anomaly detection**, reducing investigation time by 40% through structured debugging, **observability improvements**, and data-driven root cause analysis

DRONGO AI (Early-Stage AI Startup) | Bangalore, India

Machine Learning Intern | *Computer Vision & Real-Time ML Systems* **Jan 2022 – Jun 2022**

- Developed and evaluated **deep learning segmentation models** (UNet, PraNet) for medical image analysis using TensorFlow and Keras, applying **supervised learning** with data labeling, preprocessing, and evaluation pipelines that improved Dice coefficient by 10%
- Engineered **CUDA-based GPU optimization** for PyTorch training pipelines, implementing batching strategies and **parallel data loading** to reduce training time and improve **inference throughput** on large-scale image datasets, deployed on Docker, Kubernetes, and AWS EC2

PROJECT EXPERIENCE

ReconAI – Salesforce FSC Integration Observability & AI Root Cause Analysis **Mar 2026 – May 2026**

- Architected a **4-agent LangGraph hypothesis graph** for Salesforce FSC integration failure diagnosis, with **vector search** via pgvector, SSE streaming, and RCA caching, achieving 70% root cause accuracy, 12.5s average latency, and $0.006 per incident
- Built an end-to-end **fullstack AI observability platform** using FastAPI, React 18, TypeScript, and Supabase PostgreSQL vector store, replacing 3-hour manual reconciliation workflows with **AI-powered automation** delivering root cause analysis in under 15s per incident

Multi-Agent Financial Signal Intelligence System **Feb 2025 – May 2025**

- Engineered a **role-based multi-agent pipeline** using CrewAI and FinBERT embeddings that synthesizes structured financial signals from unstructured data, producing **Buy/Hold/Sell recommendations** benchmarked at BERTScore F1 > 0.82 against analyst consensus
- Designed **agentic LLM workflows** with structured output schemas and role specialization, demonstrating production-ready coordination across **agentic frameworks** for retrieval, reasoning, and signal generation in a finance domain context

TECHNICAL SKILLS

Programming & Languages: Python (Advanced); Java; JavaScript/Node.js; SQL; OOP; DSA; RESTful APIs; Microservices; Distributed Systems
AI & ML: LLMs & Generative AI; Agentic AI; RAG; FAISS; Vector Databases; Prompt Engineering; NLP; PyTorch; TensorFlow; scikit-learn; Hugging Face; Model Evaluation; Hallucination Mitigation; AI Safety; Anomaly Detection; CUDA
Backend & Infrastructure: FastAPI; Docker; Kubernetes; AWS (EC2, S3); CI/CD; PostgreSQL; Redis; Celery;Git; Cloud-Native Architecture
Data & Analytics: Snowflake; SQL Analytics; Pandas; NumPy; Feature Engineering; Statistical Analysis; Data Quality Management

EDUCATION

University of Illinois Urbana – Champaign, Champaign, Illinois **Aug 2024 - May 2026**
Master of Science Information Management, **GPA 3.93/4.00**

PES University, Bangalore, India **Aug 2018 - May 2022**
Bachelor of Technology Electronics and Communications Engineering, **GPA 7.72/10.00**